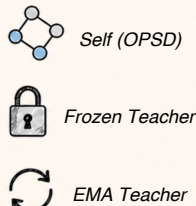
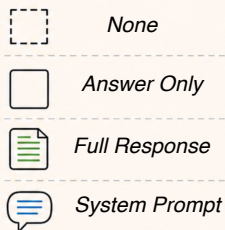


OP(S)D Design Space

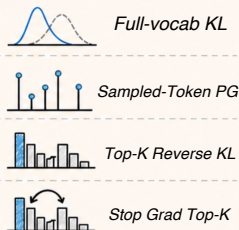
Teacher Construction



Privileged Information

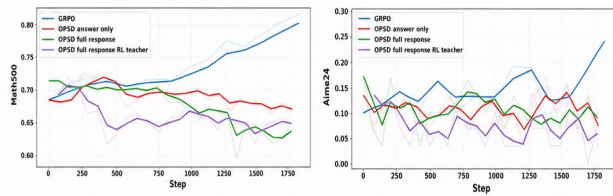
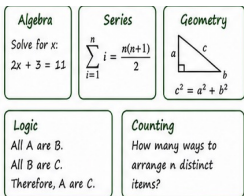


Distillation Loss Design

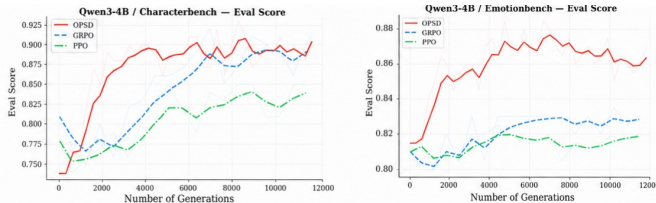
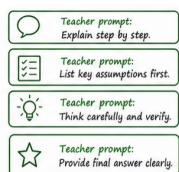


OPD Success/Failure Scenario

OPD sometimes fails in reasoning tasks.



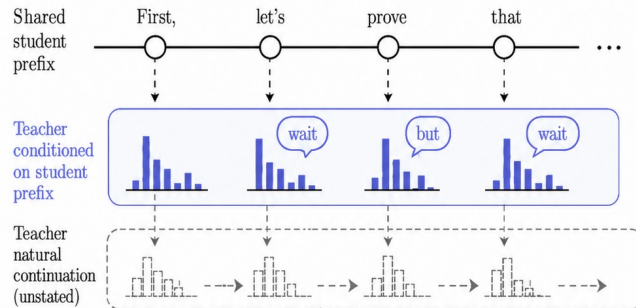
OPD succeeds in prompt internalization and style learning.



Why Does OP(S)D Fail?

1. Student prefixes distort teacher state

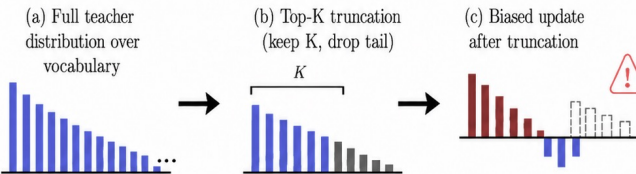
A student-generated prefix can push the teacher into an unnatural reasoning state.



Student prefixes distort the teacher state and induce revision-token supervision.

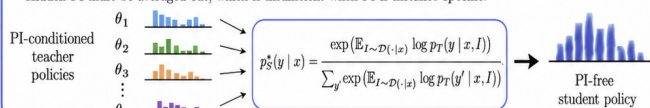
2. Unnormalized Top-K reverse-KL is unstable

Top-K truncation changes the gradient and introduces bias.



3. OPSD learns a PI-free policy

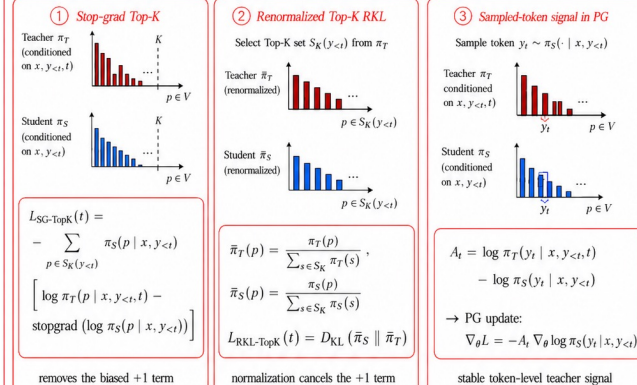
Hidden PI must be averaged out, which is insufficient when PI is instance-specific.



Improvements for OPD

1. Designing an appropriate OPD loss

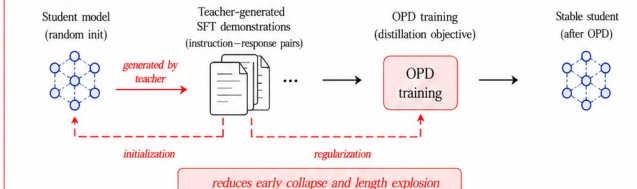
Correct the unstable unnormalized Top-K reverse-KL with stable alternatives.



All three avoid the unstable unnormalized Top-K reverse-KL update.

2. SFT stabilization

Use SFT as initialization or regularization for stable OPD.



3. RLVR-adapted teacher

Adapt the teacher on the training distribution before distillation.

